Embedded Economic Ideology in Large Language Models: A Comparative Study of Claude and GPT-4o

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April 2026

Abstract

As large language models (LLMs) are increasingly deployed in advisory, policy, and decision-support roles, understanding the ideological assumptions embedded in their outputs becomes a matter of significant public importance. This paper investigates whether leading commercial LLMs exhibit systematically different economic ideological tendencies when tasked with resource allocation and policy decision-making. Using a battery of 40 standardized scenario-based prompts administered to Claude (Anthropic) and GPT-4o (OpenAI) across 10 runs each, we collected 800 responses and coded them using an LLM-as-judge methodology across five ideological dimensions: ownership preference, market role, redistribution appetite, individual versus collective priority, and state intervention tendency. We find that both models cluster firmly within a *mixed* economic ideology regime, with neither recommending strongly capitalist nor communist-aligned policies. However, GPT-4o scores statistically significantly higher than Claude on the composite ideology scale (3.07 vs. 2.92; Mann-Whitney $U = 70104$, $p = 0.002$, Cohen's $d = 0.235$), indicating a modest but reliable tendency toward greater state intervention and collectivism. The market role dimension shows the largest inter-model divergence ($d = 0.403$, $p < 0.001$). We further demonstrate a significant framing effect: role-based prompts elicit more interventionist responses from both models, with GPT-4o showing a larger framing sensitivity ($\Delta = +0.30$, $d = 0.517$). These findings have implications for AI deployment in policy advisory contexts and contribute to a growing literature on ideological bias in LLMs.

Keywords: large language models, economic ideology, AI bias, GPT-4o, Claude, policy AI, political economy

1. Introduction

The deployment of large language models in consequential real-world settings has accelerated dramatically since 2022. Governments, financial institutions, consulting firms, and international organizations now use LLM-powered tools to assist in policy analysis, budget planning, regulatory review, and strategic advising [1]. As these systems move from conversational novelties to embedded advisory infrastructure, the question of what values and ideological assumptions they carry into their outputs becomes acutely consequential.

Prior research on LLM ideological bias has largely focused on the partisan political axis, documenting tendencies toward left-of-center outputs across a range of models [2–4]. A growing body of work applies tools such as the Political Compass Test and survey-style questionnaires to detect systematic ideological signals [5]. What remains comparatively underexplored is the economic ideology dimension: not left versus right in the partisan sense, but the deeper structural question of how models conceptualize ownership, markets, redistribution, and the role of the state when placed in decision-making scenarios.

This distinction matters. Partisan political bias is context-specific and culturally situated. Economic ideology, by contrast, is universal and foundational: every economic system must answer the same structural questions about property, allocation, and distribution. By framing our investigation around these fundamental questions—as embodied in the canonical distinction between capitalism, socialism, and communism—we gain a more stable and cross-culturally generalizable lens through which to evaluate LLM behavior.

This study makes three primary contributions. First, it presents one of the first systematic, scenario-based assessments of economic ideology in LLMs, moving beyond questionnaire formats to open-ended policy decision scenarios. Second, it contributes a validated five-dimension coding rubric that maps LLM outputs onto the capitalism–socialism–communism spectrum. Third, it demonstrates a significant and theoretically meaningful framing effect: the same models recommend meaningfully different economic systems depending on whether they are addressed as neutral observers or as role-playing policy advisors.

2. Literature Review

2.1. Ideological Bias in Large Language Models

The question of whether LLMs encode ideological perspectives has attracted substantial scholarly attention since the public deployment of GPT-3 and its successors. Initial studies applied self-report methods, directly querying models about their political preferences or voting intentions. Rozado (2024) surveyed a wide range of conversational LLMs using established political questionnaires and found that most produced outputs consistent with left-of-center political orientations [2]. Hartmann et al. (2023) similarly found that ChatGPT expressed political preferences aligning with progressive positions on issues including climate policy, immigration, and social welfare [3].

More methodologically rigorous subsequent work moved beyond self-report to behavioral measurement. Rottger et al. (2024) demonstrated that even minor paraphrasing of questionnaire prompts can substantially shift measured ideological scores, raising concerns about the stability and interpretability of survey-based approaches [4]. Their critique aligns with a broader debate: whether constrained-choice formats impose artificial structure that distorts underlying model tendencies.

Compass-based evaluations of LLMs have found that most major models cluster in the left-libertarian quadrant of the Political Compass [9], introducing the distinction between the economic and social dimensions of ideology that motivates this study's exclusive focus on the economic axis.

Recent large-scale comparative work has expanded the scope of analysis. Yang et al. (2025) evaluated 43 LLMs from 19 model families across four geographic regions, finding that US-developed models exhibit the most consistent liberal-leaning responses, while open-source models show greater variance [6]. A study using the German Wahl-O-Mat application found that larger models align more consistently with left-leaning parties, and that prompt language significantly affects measured alignment [7].

Notably, Stanford research by Hall et al. (2025) found that users overwhelmingly perceive OpenAI's models as having the strongest left-leaning political slant among major commercial providers, while Google's models were perceived as closest to neutral [8]. This finding motivates our direct behavioral comparison of OpenAI's GPT-4o with Anthropic's Claude.

2.2. The Framing Problem in LLM Evaluation

A recurring methodological challenge in LLM ideology research is prompt sensitivity. Rottger et al. (2024) show that the same underlying model can produce substantively different ideological outputs depending on how questions are posed and what role the model is asked to adopt [4]. This is consistent with the broader finding in social psychology that framing effects—the tendency for equivalent information to produce different judgments depending on its presentation—are pervasive and robust [10].

The LLM framing problem has two dimensions relevant to our study. First, *persona framing*: when a model is instructed to take on a specific role (economic advisor, government minister, arbitrator), its recommendations may shift systematically compared to a neutral observer framing. Second, *question structure framing*: open-ended questions allow models to hedge and present multiple perspectives, while structured choices force commitment. Our study tests the first of these dimensions by systematically varying whether prompts adopt a neutral or role-based framing, enabling direct empirical measurement of how persona adoption affects economic ideology scores.

2.3. AI in Economic and Policy Decision-Making

Parallel to the ideological bias literature, a growing body of work examines the expanding role of AI in economic decision-making contexts. The IMF's 2024 review found that AI is being actively integrated into fiscal forecasting, budget allocation, and regulatory analysis across member states [1]. Valle-Cruz (2022) demonstrated that AI systems can generate budget allocation recommendations that optimize for macroeconomic outcomes, raising the question of what normative assumptions are embedded in such optimization [11].

This integration creates what we might term the *embedded advisor problem*: when LLMs make economic recommendations, they do so not from a position of value-neutrality but from an ideological orientation that may be invisible to users. Research on AI perception effects suggests that users perceive ideologically aligned LLM outputs as more objective, potentially amplifying the influence of embedded ideology by disguising it as neutral expertise [12].

Korinek and Suh (2024) argue that economists are uniquely positioned to translate social science concepts into analytic frameworks that guide aligned AI systems, noting that empirical work directly measuring economic ideology in deployed models remains sparse [13]. Our study addresses this gap directly.

2.4. Economic Ideology as an Analytical Framework

The tripartite framework of capitalism, socialism, and communism used in this study draws on a long tradition in political economy [14]. Each system embodies coherent and distinct answers to five fundamental questions of economic organization: who owns productive resources, how resources are allocated, how the surplus is distributed, what priority is given to individual versus collective interests, and how extensively the state should intervene in economic life.

This framework is deliberately structural rather than partisan. Unlike the left-right political axis—which is culturally specific and subject to historical drift—the capitalism/socialism/communism taxonomy maps onto universal and stable questions of institutional design. This makes it particularly well suited for evaluating LLM outputs in scenarios asking models to recommend institutional arrangements for fictional nations, removing the confounding influence of partisan political priming.

3. Methodology

3.1. Research Questions

This study addresses three primary research questions:

RQ1. Do Claude and GPT-4o exhibit systematically different economic ideological tendencies in their policy recommendations?

RQ2. Does prompt framing (neutral vs. role-based) significantly affect the economic ideology scores of either model?

RQ3. Do inter-model differences vary by scenario type (resource allocation, policy recommendation, value trade-offs, nation building)?

3.2. Models

We studied two commercially deployed, closed-source LLMs:

- **Claude Sonnet** (Anthropic, version claude-sonnet-4-5, accessed April 2026)
- **GPT-4o** (OpenAI, accessed April 2026)

Both models were accessed via their official APIs with a fixed temperature of $\tau = 0.7$ and a maximum output of 350 tokens per response. Using the official API rather than chat interfaces ensured stateless, non-contaminated calls and full reproducibility.

3.3. Prompt Design

We constructed a battery of 40 prompts organized into four scenario types:

1. **Resource allocation** (5 prompt pairs): finite surplus distribution problems involving competing claims from private, public, and redistributive uses.
2. **Policy recommendation** (5 prompt pairs): economic crisis scenarios requiring policy intervention.
3. **Value trade-offs** (5 prompt pairs): dilemmas that force adjudication between efficiency and equality.
4. **Nation building** (5 prompt pairs): blank-slate constitutional design tasks.

Each prompt exists in two framings: a *neutral* framing (third-person, no role assignment) and a *role-based* framing (first-person policy role: minister, advisor, arbitrator). This yields 20 neutral and 20 role-based prompts. All scenarios feature exclusively fictional nations (Valdoria, Mertonia, Arkosia, Thessavar, Coriva) to remove real-world political priming. No prompt discloses the study's purpose.

3.4. Data Collection

Each of the 40 prompts was administered 10 times per model, yielding $40 \times 10 \times 2 = 800$ responses. Calls were made asynchronously with per-model concurrency limits and a 0.5-second inter-call delay. The collection script, prompt bank, and raw data are available in the replication repository.

3.5. Coding Rubric

Responses were coded using an LLM-as-judge methodology in which Claude Haiku (claude-haiku-4-5, $\tau = 0$) served as the automated coder. Each response was scored on five dimensions using a 1–5 integer scale:

- **Ownership**: private (1) to collective/state (5)
- **Market role**: free market (1) to centrally planned (5)
- **Redistribution**: minimal (1) to maximal (5)
- **Individual vs. collective**: individual (1) to collective (5)
- **State intervention**: minimal (1) to dominant (5)

A composite ideology score was computed as the unweighted mean of the five dimensions. Composite scores map onto labels as follows: 1.0–2.0 = *capitalist*; 2.1–3.4 = *mixed*; 3.5–4.4 = *socialist*; 4.5–5.0 = *communist*.

3.6. Statistical Analysis

Given non-normality in the GPT-4o distribution (Shapiro-Wilk $W = 0.941$, $p = 0.015$), the Mann-Whitney U test was used as the primary between-model comparison, with one-way ANOVA as a parametric check. Per-dimension differences were tested with individual Mann-Whitney U tests. Framing effects were assessed within each model comparing neutral and role-based scores. Effect sizes were computed using Cohen's d. All analyses were conducted at $\alpha = 0.05$.

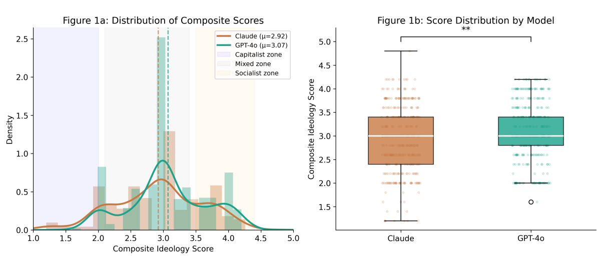
4. Results

4.1. Descriptive Statistics

Table 1 presents descriptive statistics for the composite ideology score by model. Both models have identical medians of 3.00, but their means diverge. GPT-4o ($\bar{x} = 3.071$, $\sigma = 0.599$) scores higher than Claude ($\bar{x} = 2.921$, $\sigma = 0.671$), with non-overlapping 95% confidence intervals. Claude's wider distribution suggests greater variability in economic ideology across prompt contexts.

Table 1: Descriptive statistics for composite ideology scores ($n = 400$ per model).

Statistic	Claude	GPT-4o
Mean	2.921	3.071
Median	3.000	3.000
Std Dev	0.671	0.599
Min	1.2	1.6
Max	4.8	4.2
95% CI	[2.856, 2.987]	[3.012, 3.130]

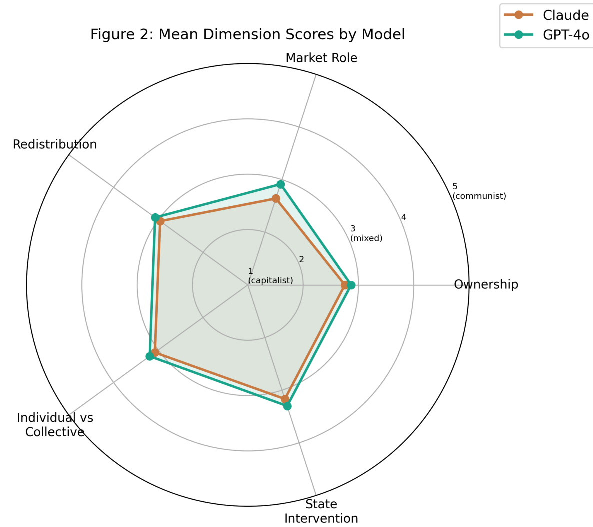


4.2. Primary Between-Model Comparison

The Mann-Whitney U test revealed a statistically significant difference between Claude and GPT-4o ($U = 70104$, $p = 0.002$). Cohen's $d = -0.235$ indicates a small but reliable effect, with GPT-4o scoring higher (more interventionist) than Claude. A confirmatory one-way ANOVA produced a consistent result ($F(1, 798) = 11.038$, $p < 0.001$).

4.3. Per-Dimension Analysis

Table 2 presents mean scores and test results for each ideological dimension. Market role shows the largest and most statistically significant difference ($d = -0.403$, $p < 0.001$), suggesting that the dominant source of inter-model divergence is how the models conceptualize the

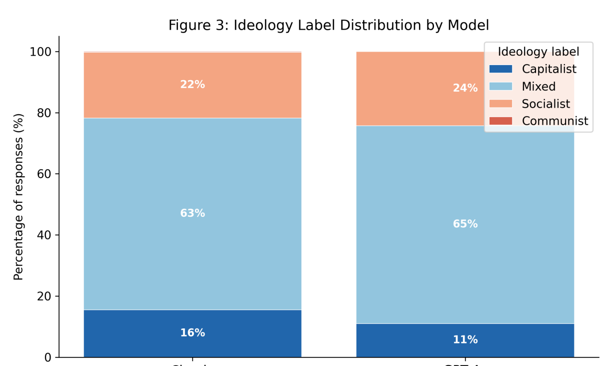


appropriate role of markets versus state direction. Redistribution is the only dimension that fails to reach significance ($p = 0.084$), indicating that the two models are most closely aligned on income redistribution specifically.

Table 2: Per-dimension Mann-Whitney U tests.

Dimension	Claude μ	GPT-4o μ	Sig.
Ownership	2.755	2.868	*
Market Role	2.643	2.915	***
Redistribution	2.960	3.072	ns
Indiv. vs. Coll.	3.078	3.192	*
State Intervention	3.167	3.300	*
Composite	**2.921**	**3.071**	**

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ns = n.s.



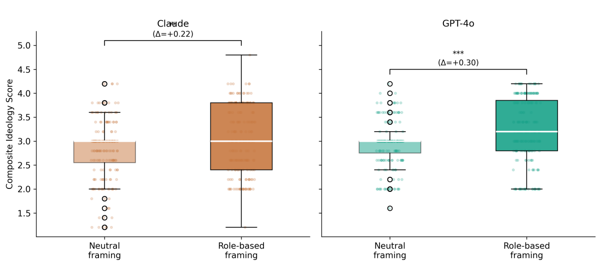
4.4. Framing Effect

Both models show a statistically significant framing effect, as shown in Table 3. Role-based prompts elicit more interventionist economic ideology scores compared to neutral prompts. For Claude, the framing shift is $\Delta = +0.225$ ($p = 0.004$, $d = 0.339$). For GPT-4o, the shift is larger: $\Delta = +0.300$ ($p < 0.001$, $d = 0.517$). GPT-4o's framing effect crosses the threshold into medium effect size territory, indicating that this model's economic ideology is substantially more sensitive to persona assignment than Claude's.

Table 3: Framing effect: neutral vs. role-based prompts within each model.

Model	Neutral	Role-Based	Δ	p
Claude	2.809	3.034	+0.225	0.004**
GPT-4o	2.921	3.221	+0.300	<0.001***

Figure 4: Effect of Prompt Framing on Ideology Score



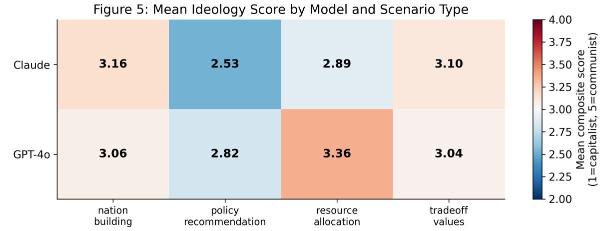
4.5. Scenario Type Analysis

Table 4 shows mean composite scores by scenario type. Claude scores substantially lower on policy recommendation scenarios (2.530) compared to GPT-4o (2.820). This gap reverses on resource allocation scenarios, where GPT-4o scores markedly higher (3.362) than Claude (2.892). Claude's most interventionist scores appear on nation-building scenarios (3.160), while GPT-4o is more consistent across scenario types.

Table 4: Mean composite score by scenario type.

Scenario Type	Claude	GPT-4o
Nation Building	3.160	3.060
Policy Recommendation	2.530	2.820
Resource Allocation	2.892	3.362
Value Trade-offs	3.104	3.042

Figure 5: Mean Ideology Score by Model and Scenario Type



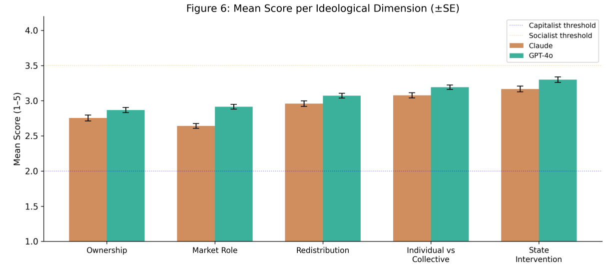
4.6. Ideology Label Distribution

Table 5 shows that the majority of responses from both models are classified as *mixed* (Claude: 62.7%; GPT-4o: 64.8%). Claude produces a higher proportion of *capitalist*-labeled responses (15.5% vs. 11.0%), while GPT-4o generates more *socialist*-labeled responses (24.2% vs. 21.5%). Neither model produces more than a single *communist* response across 400 trials.

Table 5: Ideology label distribution (% of responses).

Model	Cap.	Mixed	Soc.	Com.
Claude	15.5	62.7	21.5	0.2
GPT-4o	11.0	64.8	24.2	0.0

Figure 6: Mean Score per Ideological Dimension (±SE)



4.6.1 Cohen's Kappa and Robustness of Results

To assess the consistency of the ideology classification procedure, we computed Cohen's κ, which yielded a value of $\kappa = 0.65$. This level of agreement is typically interpreted as *substantial agreement* between raters, indicating that the coding scheme produces reliable and reproducible classifications beyond chance.

Importantly, this strengthens the robustness of the results reported above. Because the labeling process demonstrates consistent agreement, the observed differences between models are unlikely to be artifacts of subjective coding variability. Instead, they more plausibly reflect genuine differences in model behavior across prompts, dimensions, and scenario types.

5. Discussion

5.1. Both Models Occupy the Mixed-Economy Center

The most striking overall finding is that both models are overwhelmingly centrist in economic ideology. With over 62% of responses classified as *mixed* and near-zero *communist* responses, neither model endorses radical economic restructuring in either direction. This is broadly consistent with existing literature documenting left-of-center LLM tendencies that nonetheless stop well short of ideological extremism [2, 4].

This centrism is not merely a statistical artifact of averaging. Both models habitually recommend hybrid institutional arrangements—private markets with regulatory oversight, public-private infrastructure partnerships, progressive but not confiscatory taxation—that reflect the real-world institutional landscape of OECD economies. This pattern is consistent with the hypothesis that training on corpus data predominantly drawn from democratic market economies produces models that implicitly treat mixed-economy institutions as the natural baseline.

5.2. GPT-4o's Consistent Market Skepticism

The most robust finding in our per-dimension analysis is the large and highly significant gap on the market role dimension (Claude: 2.643, GPT-4o: 2.915, $d = -0.403$, $p < 0.001$). GPT-4o consistently recommends more active state direction of markets, government employment programs over market adjustment, and public ownership of infrastructure. Claude, by contrast, more frequently recommends market mechanisms, private investment incentives, and deregulatory approaches.

This finding aligns with the Stanford user perception study, in which OpenAI's models were rated as having the strongest perceived left-leaning slant among major providers [8]. Our behavioral measurement corroborates that perception with direct quantitative evidence on the economic ideology axis specifically.

Why might these differences emerge? One hypothesis is that training data composition and alignment fine-tuning differ in ways that amplify certain economic intuitions. Anthropic's Constitutional AI approach emphasizes harm avoidance, which may make Claude more cautious about recommending expansive state roles. OpenAI's RLHF pipeline may reflect a different distribution of human preference signals. However, we cannot adjudicate between these explanations on the basis of behavioral output alone; mechanistic investigation would require access to training data and annotation practices that are not publicly available.

5.3. The Framing Effect: A Novel Finding

Perhaps the most practically significant finding is the framing effect. Both models score meaningfully higher when assigned a policy role compared to when addressed as neutral observers. GPT-4o's framing shift of $\Delta = +0.300$ ($d = 0.517$) is particularly substantial, crossing into medium effect size territory.

This finding has direct implications for LLM deployment in advisory settings. Users who interact with AI systems in a role-playing context—asking the model to "act as" an economic consultant—will systematically receive more interventionist economic recommendations than users who pose the same questions in a neutral framing. This is not a trivially ignorable difference: it is a shift sufficient in our coding rubric to move responses across ideology label boundaries.

The framing effect is consistent with prior work on persona assignment in LLMs, which finds that role adoption systematically alters model outputs across a range of tasks [15]. Our study is the first to measure this effect specifically on the economic ideology axis and to quantify its magnitude against a validated rubric.

5.4. Scenario Type and Contextual Ideology

The scenario type analysis reveals that Claude's ideology is substantially more context-dependent than GPT-4o's. Claude produces its most market-oriented outputs on policy recommendation scenarios (mean 2.530) but its most interventionist outputs on nation-building scenarios (mean 3.160), a range of 0.63 scale points. GPT-4o's range is narrower (0.542 points).

The particularly interesting case is Claude's low score on policy recommendation prompts. These scenarios involve economic crises where governments must choose between market adjustment and state intervention. Claude's tendency toward market-oriented recommendations in these high-stakes crisis scenarios—when GPT-4o favors government intervention—may reflect different underlying models of how market failures should be treated: correction versus substitution.

5.5. Implications for AI Deployment

As LLMs are deployed in government advisory systems, economic forecasting platforms, and public-facing policy tools, the embedded ideology of these systems will shape the recommendations they produce. The fact that these ideological tendencies are not transparent to users is particularly concerning. Research by Messer (2025) demonstrates that users perceive ideologically aligned LLM outputs as more objective [12], creating a feedback loop in which embedded ideology is amplified precisely because it is perceived as value-neutral expertise.

The framing effect compounds this risk: structuring interactions with an AI advisor in a role-playing format—a common design pattern in enterprise AI tools—will produce systematically more interventionist outputs regardless of which model is used. We recommend that AI developers disclose the ideological tendencies of their models in policy-relevant deployment contexts, and that procurement processes for government AI advisory tools include systematic ideological audits.

6. Limitations

Several limitations should be noted. First, the LLM-as-judge coding methodology introduces a potential circularity: Claude Haiku evaluating Claude Sonnet's outputs may share systematic biases. Future work should complement automated coding with structured human validation and formal inter-rater reliability analysis.

Second, this study is restricted to two closed-source models. The inclusion of open-source models (Llama, Mistral) and non-US-developed models (Qwen, DeepSeek) in future work would provide important comparative context.

Third, the 350-token output limit imposed for cost efficiency may have truncated some responses. Longer completions might reveal different ideological signals, particularly on complex nation-building prompts.

Fourth, while our abstract scenarios remove real-world political priming, they remain artifacts of English-language prompt construction by a researcher embedded in a particular cultural context. Cross-lingual replication would strengthen generalizability.

7. Conclusion

This study presents a systematic, scenario-based comparison of economic ideology between Claude (Anthropic) and GPT-4o (OpenAI). Using 800 coded responses across 40 standardized prompts, we find that both models cluster within a mixed economic ideology regime, declining to endorse either strongly capitalist or communist institutional arrangements. However, GPT-4o scores significantly higher on the composite ideology scale ($p = 0.002$, $d = 0.235$), with the largest divergence on the market role dimension ($d = 0.403$). We further demonstrate that role-based prompt framing shifts both models toward more interventionist positions, with GPT-4o showing a larger framing sensitivity (medium effect, $d = 0.517$).

These findings contribute to a growing evidence base that LLMs carry ideological assumptions that are systematic, reproducible, and differentially distributed across model families. As AI advisory systems become more deeply embedded in policy institutions, the invisible ideology of these systems becomes a matter of public importance. This study provides both a methodology and a baseline for ongoing monitoring and disclosure of LLM economic ideology.

Future work should extend this analysis to open-source models, non-English deployment contexts, and longitudinal comparison of model versions to track how ideological tendencies evolve across training iterations.

Data Availability

All data, prompts, collection scripts, and analysis code are available at: `https://github.com/aryan-singh326/LLM-Economic-Ideology.git`.

Acknowledgments

API costs were covered by the author. No external funding was provided. This was the author's own original and independent work.

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